



Launch a Kubernetes Cluster



Nabil Ahmed

Nodes (3) Info							
Filter Nodes by property or value							
Node name	Instance type	Compute	Managed by	Created	Status	CPU usage	
ip-192-168-16-190.eu-west-2.compute.internal	t3.micro	Node group	nextwork-nodegroup	17 minutes ago	Ready	17%	
ip-192-168-38-0.eu-west-2.compute.internal	t3.micro	Node group	nextwork-nodegroup	17 minutes ago	Ready	17%	
ip-192-168-75-213.eu-west-2.compute.internal	t3.micro	Node group	nextwork-nodegroup	17 minutes ago	Ready	22%	



Introducing Today's Project!

In this project, I will deploy my very first Kubernetes cluster using Amazon EKS. This is because EKS is AWS' service for deploying Kubernetes clusters in the cloud. I will also learn about tools like `eksctl` and CloudFormation.

What is Kubernetes?

Kubernetes is. a tool for automating the management of running containers/containerised applications, also known as container orchestration. Companies and developers use Kubernetes to deploy and manage large scale containerised applications.

I used `eksctl` to create a Kubernetes cluster using the command line. The create cluster command I ran defined the name of the cluster, its node group's name and node size settings. I also defined the region and instance type of EC2.

I initially ran into two errors while using eksctl. The first one was because of not having eksctl installed. The second one was because the EC2 instance did not have permissions to my AWS account and services yet.

```

#
' \ ##\# Amazon Linux 2023
-- \ ##\#
-- \ ##\#
-- \# /
-- v- ' ->
--
-- .-
-- /m/ '
[ec2-user@ip-172-31-3-7 ~]$ eksctl create cluster \
--name nextwork-eks-cluster \
--nodegroup-name nextwork-nodegroup \
--node-type t3.micro \
--nodes 3 \
--nodes-min 1 \
--nodes-max 3 \
--version 1.33
-bash: eksctl: command not found

```



eksctl and CloudFormation

CloudFormation helped create my EKS cluster because eksctl uses CloudFormation under the hood to actually create the cluster when I ran an eksctl create command. It created VPC resources because creating the EKS cluster in my default VPC would cause compatibility and permission issues.

There was also a second CloudFormation stack for the node group. The difference between a cluster and node group is that the cluster is the entire Kubernetes setup (including the control plane), while the node group is just a group of EC2 instances inside.

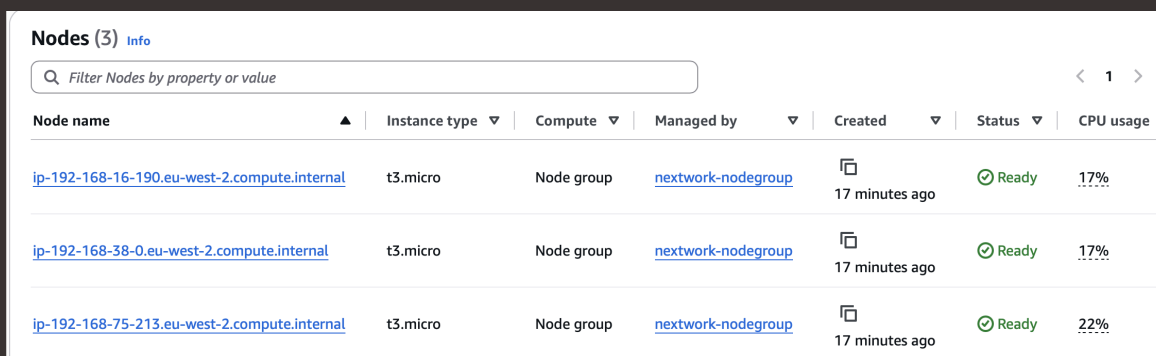
Stacks	
☐	eksctl-nextwork-eks-cluster-nodegroup-nextwork-nodegroup 2026-07-13 10:27:02 UTC+0100 ✔ CREATE_COMPLETE
☒	eksctl-nextwork-eks-cluster-cluster 2026-07-13 10:15:59 UTC+0100 ✔ CREATE_COMPLETE



The EKS console

I had to create an IAM access entry in order to see the nodes in my new node group. An access entry is a mapping of AWS IAM policies to Kubernetes' access control system. I set it up by using the Access Entries page within the EKS management console.

It took 20 minutes to create my cluster. Since I'll create this cluster again in the next project of this series, maybe this process could be sped up if I used templates e.g. Terraform or CloudFormation, and install dependencies ahead.



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